

Tutorial – Week 12

White-box Testing

```
binarySearch(elem searchKey, elem* array, int arraySize, bool &found, int &index)
{
    int bot, mid, top;                                (1)
    index = -1;
    bot = 0;
    top = arraySize - 1;
    mid = (top + bot)/2;

    if (array[mid] == searchKey)                       (2)
        found = true;                                 (3)
    else
        found = false;                                (4)

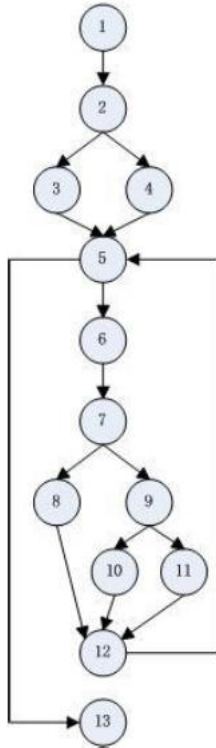
    while (bot <= top && !found)                       (5)
    {
        mid = (top + bot)/2;                           (6)
        if (array[mid] == searchKey)                   (7)
        {
            found = true;                               (8)
            index = mid;
        }
        else if (array[mid] < searchKey)                (9)
            bot = mid + 1;                             (10)
        else
            top = mid - 1;                             (11)
    }                                                    (12)
}                                                        (13)
```

The above is a binary search function written in C language. Answer the following questions base on the code.

Question 1

Draw a control flow graph corresponding to the preceding function.

Answer



Question 2

For the preceding function, derive a set of test cases that will cover branch testing.

Answer

	Path	Test Case
P1	1-2-3-5-13	arraysize = 3, searchkey = b, array = abc
P2	1-2-4-5-13	arraysize = 0, searchkey = a
P3	1-2-4-5-6-7-8-12-5-13	impossible
P4	1-2-4-5-6-7-9-10-12-5-6-7-8-12-5-13	arraysize = 3, searchkey = a, array = abc
P5	1-2-4-5-6-7-9-11-12-5-6-7-8-12-5-13	arraysize = 3, searchkey = c, array = abc

Question 3

Design additional test cases that will cover multiple-condition testing.

Answer

According to *while (bot <= top && !found)*, Multiple-condition testing is as follows

bot <= top	! found	Test Case
T	T	arraysize = 3, searchkey = a, array = abc
T	F	arraysize = 3, searchkey = b, array = abc
F	T	arraysize = 3, searchkey = d, array = abc
F	F	impossible

Question 4

Determine the cyclomatic complexity of the preceding function.

Answer

$$V(G) = e - n + 2p = 16 - 13 + 2 = 5$$

Question 5

Derive a set of test cases that cover McCabe's Basis Paths.

Answer

	Path	Test Case
P1	1-2-4-5-6-7-9-10-12-5-6-7-9-11-12-5-6-7-8-12-5-13	arraysize = 7, searchkey = e, array = abcdefg
P2: Flip at 2	1-2-3-5-6-7-9-10-12-5-6-7-9-11-12-5-6-7-8-12-5-13	impossible
P3: Flip at 5	1-2-4-5-13	arraysize = 0, searchkey = a
P4: Flip at 7	1-2-4-5-6-7-8-12-5-6-7-9-11-12-5-6-7-8-12-5-13	impossible
P5: Flip at 9	1-2-4-5-6-7-9-11-12-5-6-7-9-11-12-5-6-7-8-12-5-13	arraysize = 7, searchkey = a, array = abcdefg